Weston Pangle

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EDUCATION

University Of Michigan, Ann Arbor, MI *2021 - 2025*

Bachelor's of Science in Engineering, *Computer Science Engineering*

Relevant Coursework: DS&A, Computer Organization, Software Engineering, Web Systems, Operating Systems (6 credit), Computer Networks, Game Design & Dev, Intro Machine Learning, Distributed Systems, Database Management Systems, Computer Security, Calculus (I, II, & IV), Physics (I & II), Statistics

Languages and Technologies: (Proficient) C/C++, Python, Go, Git (Intermediate) SQL, Unity, JIRA (Familiar) AWS, Docker, JavaScript, CSS, HTML, Java, MongoDB

Skills: Organization, Leadership, Communication, Problem-Solving, Teamwork, Time Management, Rapid Learning

LEADERSHIP AND EXPERIENCE

Orientation Leader – *University of Michigan* *April 2024 – Current*

Technical Game Designer – *Driver's Seat Studios: University of Michigan* *October 2024 – December 2024*

- Formulated and implemented dynamic level layouts, enhancing gameplay mechanics and user experience to create **engaging** and interactive environments
- Sourced and integrated assets to enhance visual appeal and functionality, including advanced **particle effects**, fire and water shaders, and stylized cel shading using **custom Shader Graphs**
- Constructed interactive object scripts to drive gameplay and received recognition for outstanding game design, contributing to team's **1st place** win at game showcase among **14 competing games**

Software Engineer – *Michigan Data Science Team: University of Michigan* *September 2022 – January 2023*

- Developed a Pokémon "auto-battler" using **Python** and **supervised learning techniques**, achieving an **85% success rate** in battle predictions
- Engineered an algorithm for move recommendations based on **Pokémon types, moves, and traits** to optimize competitive performance
- **Collaborated** with a team to design, train, and refine machine learning models for strategic battle simulations

PERSONAL PROJECTS

Sharded Key-Value Service w/ Paxos Groups | Distributed Systems | Go *April 2025*

- Built a distributed, fault-tolerant key/value store using Paxos-based replication and dynamic sharding across replica groups
- Designed and implemented a Shard Master service for consistent, serialized reconfiguration, enabling reliable shard reassignment during group joins and failures
- Achieved linearizable client operations with at-most-once semantics, stateful shard migration, and efficient handling of concurrent reconfigurations

Reliable Transport Protocol | Computer Networks | C++ *November 2024*

- Developed a custom reliable transport protocol over **UDP** to ensure in-order, reliable delivery of data with handling for packet loss, delay, corruption, duplication, and reordering
- Implemented a sliding window mechanism for **wSender** and **wReceiver** to manage transmission and acknowledgment of packets, including features such as retransmission timers and checksum validation
- Optimized data transfer efficiency by modifying **wSender** and **wReceiver** to handle selective acknowledgment and retransmission, reducing unnecessary data retransmissions and improving network performance

Network File System | Advanced Operating Systems | C++ *April 2024*

- Engineered a multithreaded network file system, utilizing a **manager-worker** threading model to optimize handling of concurrent file operations and increase system throughput
- Devised and implemented thread-safe file and directory operations leveraging **mutexes** and **upgradeable locks**, ensuring high data integrity and minimizing bottlenecks during simultaneous access
- Tested and validated system performance across different operational states, ensuring compatibility with both **initialized** and **new file systems**, while maintaining stability and reliability under heavy client load

ADDITIONAL

More Projects: HTTP Proxy w/ DNS, Thread Library, Instagram Clone, Search Engine, Metroid Remake, SQL & NoSQL Database